GENERATIVE AI

Generative Artificial Intelligence

Prepared By

Amritanshu Singh

October 2026

Contents

1 What is GenAI? 1

1.1 Evolution of Generative AI 1

2 Five Main Types of Generative AI 2

2.1 Generative Adversarial Networks (GANs) 2

2.2 Variational Autoencoders (VAEs) 3

2.3 Autoregressive Models 4

2.4 Recurrent Neural Networks (RNNs) 6

2.5 Diffusion Models 6

3 Conclusion 8

1 What is GenAI?

Generative artificial intelligence, or GenAI, uses sophisticated algorithms to learn patterns from large and complex datasets and create new content, including text, images, audio, video, and code, in response to a query or prompt. This technology is changing enterprise technology and transforming how businesses operate.

Generative AI typically learns representations of existing information and uses the patterns it has learned to generate new content. Many models represent relationships between data points in a mathematical space. When prompted, the model uses these learned relationships to produce a suitable response.

Popular interfaces such as OpenAI's ChatGPT and Google's Gemini allow users to ask complex questions, summarize large amounts of information, generate content, and automate tasks that previously required substantial human effort.

Businesses use generative AI to:

- Draft reports, documents, and business communications.
- Personalize marketing campaigns.
- Create images, videos, and other creative content.
- Generate, explain, and improve computer code.
- Support customer service and business decision-making.
- Generate synthetic data for training and testing AI systems.

Software vendors are integrating generative AI into business applications such as Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) systems. It is also being added to automation technologies, including Robotic Process Automation (RPA) and customer service chatbots, to improve their capabilities.

1.1 Evolution of Generative AI

The foundations of generative AI date back several decades. Early chatbot systems appeared in the 1960s, while significant progress in machine learning subsequently enabled systems to generate increasingly realistic content.

In 2014, Generative Adversarial Networks (GANs) introduced an important approach for producing realistic synthetic data. Later advances in transformer architectures and large language models (LLMs) significantly improved the ability of AI systems to process context and generate coherent content.

Transformers use a mechanism called attention to model relationships between different parts of an input. This enables models to capture connections between words and passages across long documents. The same general architecture can also be adapted to tasks involving computer code, biological sequences, chemical structures, and other types of data.

The public launch of ChatGPT in late 2022 brought generative AI to widespread public attention by making conversational AI and rapid text generation accessible to a broad audience.

2 Five Main Types of Generative AI

Generative AI includes several model families, each designed to learn patterns from data and generate new outputs. The following sections explain five important types, their working principles, and their practical applications.

2.1 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are generative models consisting of two main components: a generator and a discriminator. These components are trained in competition with each other.

Generator: Produces new data samples based on patterns learned from the training dataset.

Discriminator: Evaluates whether a sample appears to come from the real dataset or has been generated artificially. Its feedback helps the generator improve its outputs.

Through this adversarial training process, GANs can produce synthetic data that resembles real-world examples.

Analogy

Imagine a counterfeiter attempting to make fake money and a police officer attempting to identify counterfeit notes. The counterfeiter continuously improves the notes, while the officer becomes better at identifying flaws. This competition resembles the relationship between the generator and discriminator.

Use Cases

- **Financial services:** Generating synthetic transaction data for fraud-detection training without directly exposing real customer records.
- **Public sector:** Creating synthetic population datasets for policy modelling while reducing the use of identifiable personal information.
- **Manufacturing:** Producing virtual product prototypes to support design and stress testing.

Healthcare Example

A hospital wants to improve its lung cancer detection system but has only a limited number of high-quality CT scans. Data scientists could use GANs to generate synthetic CT images representing lungs with and without tumours.

These synthetic images may supplement the training dataset of a diagnostic AI model. If the generated images are realistic and clinically validated, they can help address data shortages and support model development while reducing reliance on additional patient scans. Synthetic data must still be evaluated carefully because it may contain inaccuracies or fail to represent important clinical cases.

2.2 Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs) are generative models that learn compact probabilistic representations of data. They encode an input into a latent space and decode a sampled representation to reconstruct the input or generate a similar example.

Encoder: Maps the input data to a probability distribution in a lower-dimensional latent space.

Decoder: Uses a sampled latent representation to reconstruct data or generate new examples.

Analogy

Imagine a recipe book. Instead of copying every dish exactly, you record its essential ingredients and preparation steps. Later, you can vary the spices or quantities to create a slightly different dish that still resembles the original.

Similarly, a VAE learns important characteristics of the data and can use its latent representation to generate variations.

Use Cases

- **Security:** Simulating unusual cyberattack scenarios to support the training of detection systems.
- **Language and media:** Generating variations of written or spoken content for different applications.
- **Manufacturing:** Exploring alternative product designs before building physical prototypes.

2.3 Autoregressive Models

Autoregressive models generate sequences by predicting the next element based on the elements that came before it. They are widely used for text generation and can also be applied to audio, music, and time-series data.

These models learn conditional probability distributions that estimate which element is likely to come next, given the preceding context. Many large language models, including the models used in conversational systems such as ChatGPT, use autoregressive generation.

How Autoregressive Models Work

a. The model generates a sequence one element or token at a time rather than producing the entire sequence at once.

b. Each new token is predicted using the preceding context.

c. The generated token is added to the sequence and becomes part of the context used to predict the next token.

Example

Suppose the input is:

"The sun was setting. . . "

The model might predict "and". The resulting sequence becomes "The sun was setting and". It may then predict "the", followed by additional words until a complete response is generated.

Use Cases

- **Language models:** Generating text, conversational responses, and translations.
- **Finance:** Simulating plausible future price sequences for risk analysis.
- **Security and compliance:** Generating simulated phishing messages for employee awareness training.
- **Entertainment:** Composing music, generating scripts, and assisting with creative writing.

Financial Example: Stock Price Simulation

Suppose a trading firm wants to analyse how a stock might behave over the next 30 days. The firm provides the model with historical stock prices, perhaps covering the previous three years.

The model learns patterns and relationships in the historical sequence. Starting with the latest observed price, it generates a possible next-day value. That value is then used as context for generating the following day's value.

This process continues until a complete 30-day sequence has been generated. The firm can repeat the simulation thousands of times to produce multiple possible future scenarios.

Why it matters: Analysts can examine scenarios involving steady growth, sharp declines, or sudden rallies and use them to stress-test portfolios. These simulations are

not guaranteed predictions; their quality depends on the model, its assumptions, and the data available.

2.4 Recurrent Neural Networks (RNNs)

Recurrent Neural Networks (RNNs) are neural networks designed to process sequential data. They maintain a hidden state that carries information from earlier inputs into later processing steps.

This makes RNNs useful for tasks in which the order of information and its preceding context are important. Traditional RNNs can, however, have difficulty learning very long-range dependencies. Variants such as Long Short-Term Memory (LSTM) networks were developed to help address this limitation.

Use Cases

- **Language:** Generating text or speech while considering previously processed information.
- **Content generation:** Producing poetry, stories, and other sequential content.
- **Security:** Modelling network traffic sequences and helping identify unusual behaviour.

Example: RNNs in the Energy Sector

An energy provider can use recurrent neural networks to estimate electricity demand at different times of the day. The model learns from historical consumption, weather conditions, and time-related patterns.

These forecasts can help the provider balance electricity supply and demand, improve grid management, and reduce the risk of shortages. Actual results depend on data quality and the performance of the forecasting system.

2.5 Diffusion Models

Diffusion models are generative models commonly used to create images and other forms of content. They learn to reverse a process in which noise is gradually added to training data.

Imagine a clear photograph. Tiny amounts of visual noise are gradually added until the original image becomes difficult to recognize. A diffusion model learns how to reverse this corruption process.

During generation, the model typically begins with random noise and progressively transforms it into a new sample. When guided by a text prompt, it can generate images that correspond to the requested description.

Use Cases

- **Media and marketing:** Generating images for advertising campaigns and creative projects.
- **Design:** Exploring visual concepts and producing alternative design ideas.
- **Digital content:** Creating artwork, illustrations, and realistic-looking scenes.

Examples of tools that use diffusion-based image generation include Stable Diffusion and certain image-generation systems in the DALL-E family.

Design Workflow Example

Generative AI can support user experience (UX) and user interface (UI) design by helping teams explore ideas and produce visual concepts. Tools such as UX Pilot can assist designers in developing ideas based on user personas and design requirements.

Generative AI may help design teams by:

- Answering questions and suggesting creative directions.
- Supporting decisions based on user personas.
- Reducing the need to create every design from scratch.
- Generating visual concepts that follow design requirements.
- Saving time during early design exploration.

AI-generated designs still require human review to ensure usability, accessibility, consistency, and accuracy.

3 Conclusion

Generative AI is an important development in artificial intelligence that enables machines to create new content by learning patterns from existing data. It has applications in healthcare, finance, manufacturing, security, energy, entertainment, and digital design.

The five model families discussed in this report are Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), autoregressive models, Recurrent Neural Networks (RNNs), and diffusion models. Each has different strengths and is suitable for different tasks.

Generative AI offers opportunities to improve productivity, support creativity, generate synthetic data, and assist decision-making. However, its outputs must be evaluated for accuracy, bias, privacy, security, and reliability. Combining these technologies with appropriate human oversight can help organizations use generative AI responsibly and effectively.